AYUSH RANJAN

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PROFESSIONAL SUMMARY

B.Tech Computer Science Engineering graduate (2026) and AI/ML Engineer with hands-on experience building end-to-end Generative AI applications, RAG pipelines, and autonomous AI agents. Skilled in LLM integration, prompt engineering, vector databases, and MLOps, with a strong foundation in Python, machine learning, deep learning, and NLP. Proven track record deploying scalable AI solutions using FastAPI and LangGraph, and translating business requirements into production-ready, responsible AI systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL

Generative AI: LLM, Prompt Engineering, RAG, AI Agents, Vector DB, MCP

Data Science & ML: Data Analysis, Statistical Modeling, Machine Learning, Deep Learning, NLP, Computer Vision

Frameworks & Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, LangChain, LangGraph, Hugging Face

Backend & APIs: FastAPI, REST API Design, JWT Authentication & Authorization, Pydantic, Pytest

Cloud, MLOps & Tools: Excel, Tableau, Git/GitHub, Docker, Linux, CI/CD Pipelines

WORK EXPERIENCE

Trainee AI Engineer | *LevelShift* - Chennai, India — Jul 2026 – Present

- Built Retrieval-Augmented Generation (RAG) pipelines to power intelligent, context-aware application workflows.
- Developed and deployed autonomous AI agents to automate key business processes and decision-making tasks.
- Integrated third-party and internal APIs into production systems to extend functionality and streamline data flow.
- Implemented LLM cost optimization strategies, improving operational efficiency and reducing inference spend.

Machine Learning Engineer Intern | *Startcorn* - India (Remote) — Sep 2025 – Mar 2026

- Trained and evaluated Machine Learning and Deep Learning models to improve accuracy, performance, and reliability.
- Built Retrieval-Augmented Generation (RAG) pipelines for developing intelligent, context-aware AI applications.
- Developed AI agents capable of automating tasks, reasoning through multi-step workflows, and interacting with external tools and systems.
- Implemented AI guardrails to enhance the safety, reliability, and responsible use of AI systems.
- Designed and developed intelligent chatbots using Machine Learning, Generative AI, and Large Language Models (LLMs).

Prompt Engineer | *Outlier AI* - Remote — Sep 2024 – Nov 2024

- Designed, evaluated, and iteratively refined 100+ LLM prompts to improve output relevance, factual accuracy, and response quality for enterprise AI automation workflows.
- Conducted systematic benchmarking and quality evaluation of large language model responses across diverse NLP tasks.
- Developed reusable prompt templates adopted across multiple enterprise AI projects, reducing prompt iteration cycles by 40%.

PROJECTS

AI Agent for Daily Water Intake Tracking — *Python, FastAPI, LangChain, Mistral AI, SQLite*

- Developed an AI-powered hydration tracking system using FastAPI, LangChain, and Mistral AI to monitor consumption and deliver personalized, LLM-generated insights.
- Built RESTful API endpoints for logging intake, retrieving user history, and generating real-time AI-based analysis.
- Integrated SQLite for persistent storage, implemented structured logging for monitoring, and applied prompt engineering to improve recommendation quality.

Garbage Image Classifier — *Python, TensorFlow, CNN, OpenCV*

- Developed a Convolutional Neural Network (CNN) achieving 91% classification accuracy on 3,000+ labeled waste images.
- Applied data augmentation techniques (flipping, rotation, zoom) to improve model robustness and generalization.
- Implemented a real-time camera simulation pipeline using OpenCV for live inference demonstration.

Sentence Autocompletion (Next-Word Prediction) — *Python, LSTM, TensorFlow, NLP*

- Built an LSTM-based sequence model trained on 50,000+ sentences, achieving 87% next-word prediction accuracy.
- Executed an end-to-end NLP pipeline: data preprocessing, tokenization, padding, model training, and hyperparameter tuning.
- Optimized model performance through learning rate scheduling and dropout regularization.

EDUCATION

B.Tech, Computer Science & Engineering — Jul 2022 – Jun 2026

Lovely Professional University, Punjab

Senior Secondary (CBSE – 12th) — 2021

Sarala Birla Public School, Ranchi, Jharkhand